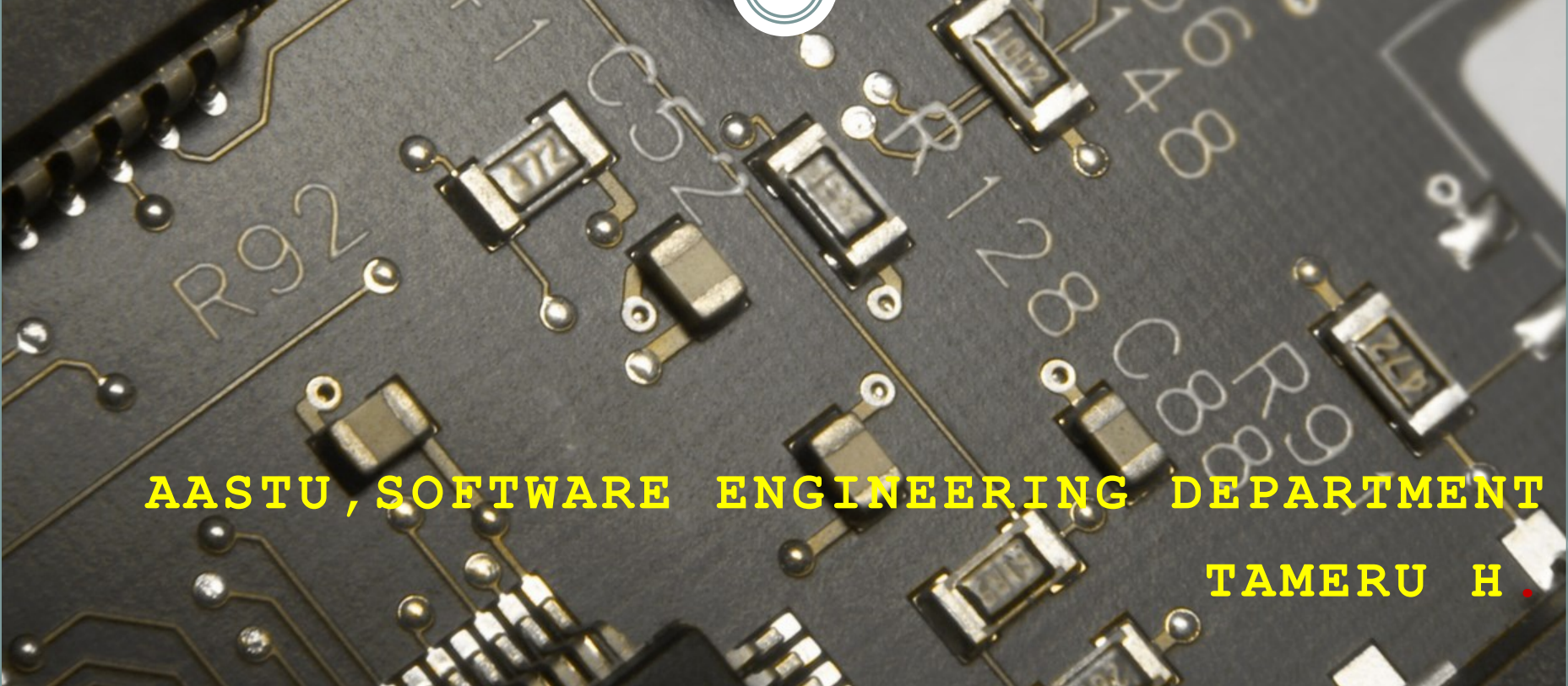


Chapter Two

Part-1

Computer Evolution



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Computer Generations

First Generation: Vacuum Tubes



Vacuum tubes: slow, expensive, fragile

Transistors: much simpler, much smaller, much cheaper, more reliable, no warm up, much faster.



Integrated circuits: miniaturization added to all the existing benefits, enabled unthought-of possibilities

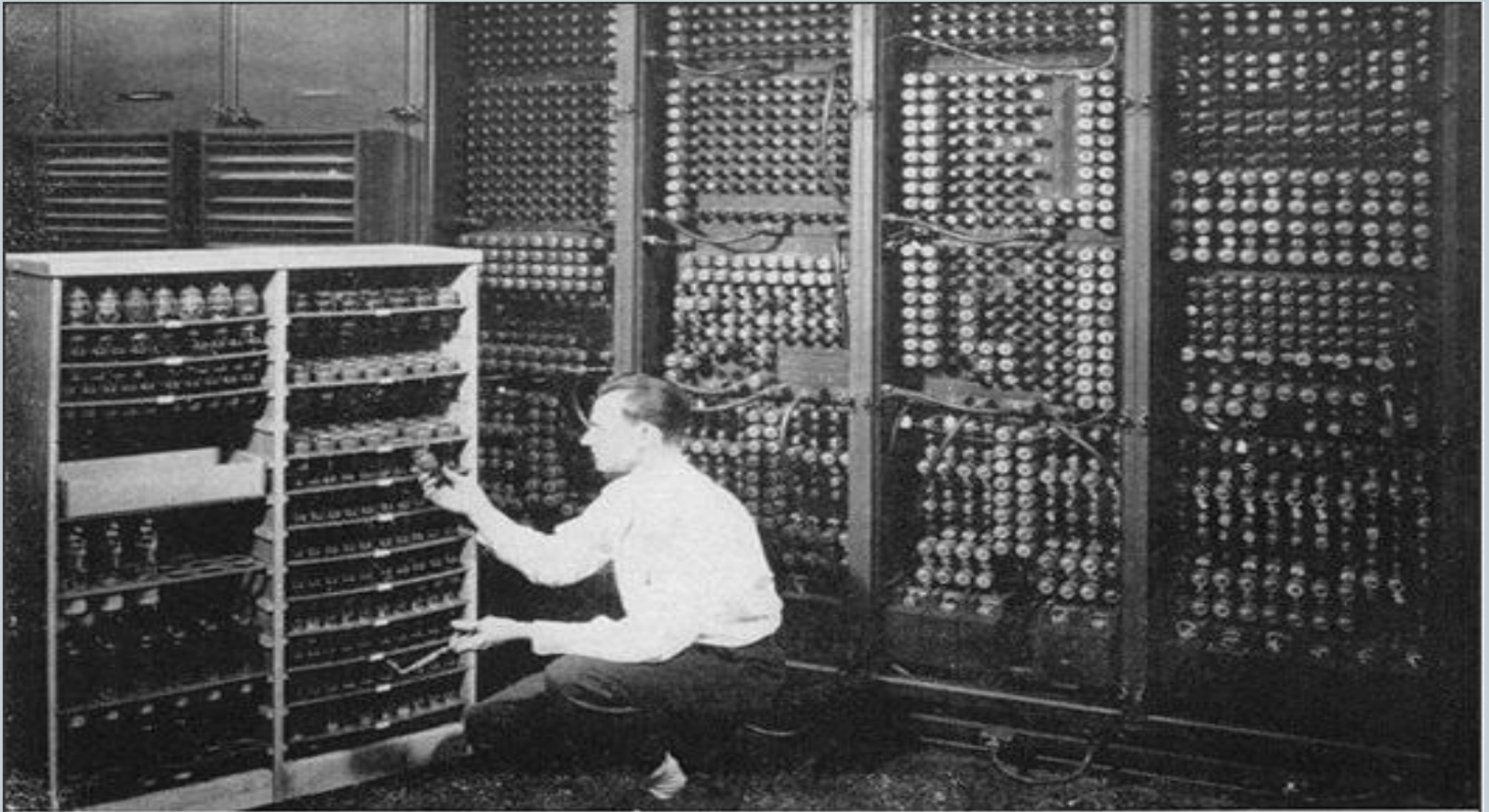
ENIAC

(Electronic Numerical Integrator And Computer)



- ❑ **ENIAC** (Electronic Numerical Integrator And Computer), designed and constructed at the University of Pennsylvania.
- ❑ The world's first general purpose electronic digital computer.
- ❑ Machine was enormous, weighing **30** tons, occupying **1500** square feet of floor space, and containing more than **18,000** vacuum tubes.
- ❑ The ENIAC was a decimal rather than a binary machine.

Cont'd...



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

Credit: U.S. Army photo

ENIAC chief developer



Credit: Reuters

❑ **John Mauchly**, a professor of Electrical Engineering at the University of Pennsylvania,

Six women who programmed ENIAC



THE VON NEUMANN MACHINE



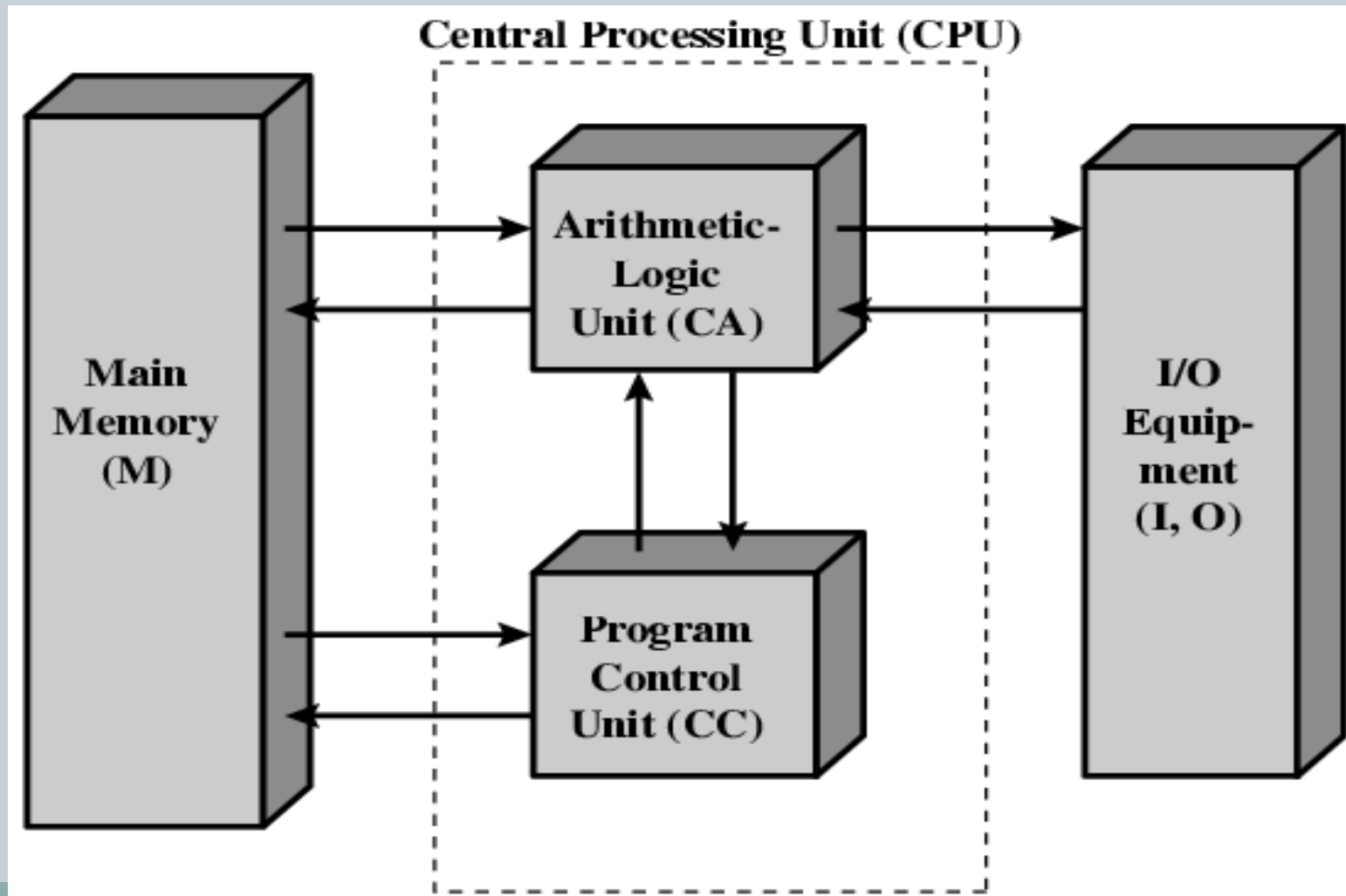
- ❑ Programing the ENIAC was extremely tedious.
- ❑ But suppose a program could be represented in a form suitable for storing **in memory** alongside the data.
- ❑ Then, a computer could get its instructions by reading them from memory, and a program could be set or altered by setting the values of a portion of memory.
- ❑ This idea, known as the **stored-program concept**,

Cont'd...



- ❑ In 1946, von Neumann the mathematician and his colleagues began the design of a new stored-program computer, referred to as the **IAS** computer, at the Princeton Institute for Advanced Studies.
- ❑ All of today's computers have this same general structure and function and are thus referred to as **von Neumann machines**.

Structure of Von Neumann machine



IAS - details



- ❑ 1000 x 40 bit words
 - ✓ Binary number
 - ✓ 2 x 20 bit instructions

- ❑ Set of registers (**storage in CPU**)
 - ✓ **Memory buffer register (MBR)**: Contains a word to be stored in memory or sent to the I/O unit, or is used to receive a word from memory or from the I/O unit.

 - ✓ **Memory address register (MAR)**: Specifies the **address** in memory of the word to be written from or read into the MBR.

Cont'd...



- ✓ **Instruction register (IR):** Contains the **8-bit opcode** instruction being executed.
- ✓ **Instruction buffer register (IBR):** Employed to hold temporarily the **right hand instruction** from a word in memory.
- ✓ **Program counter (PC):** Contains the address of **the next instruction** pair to be fetched from memory.

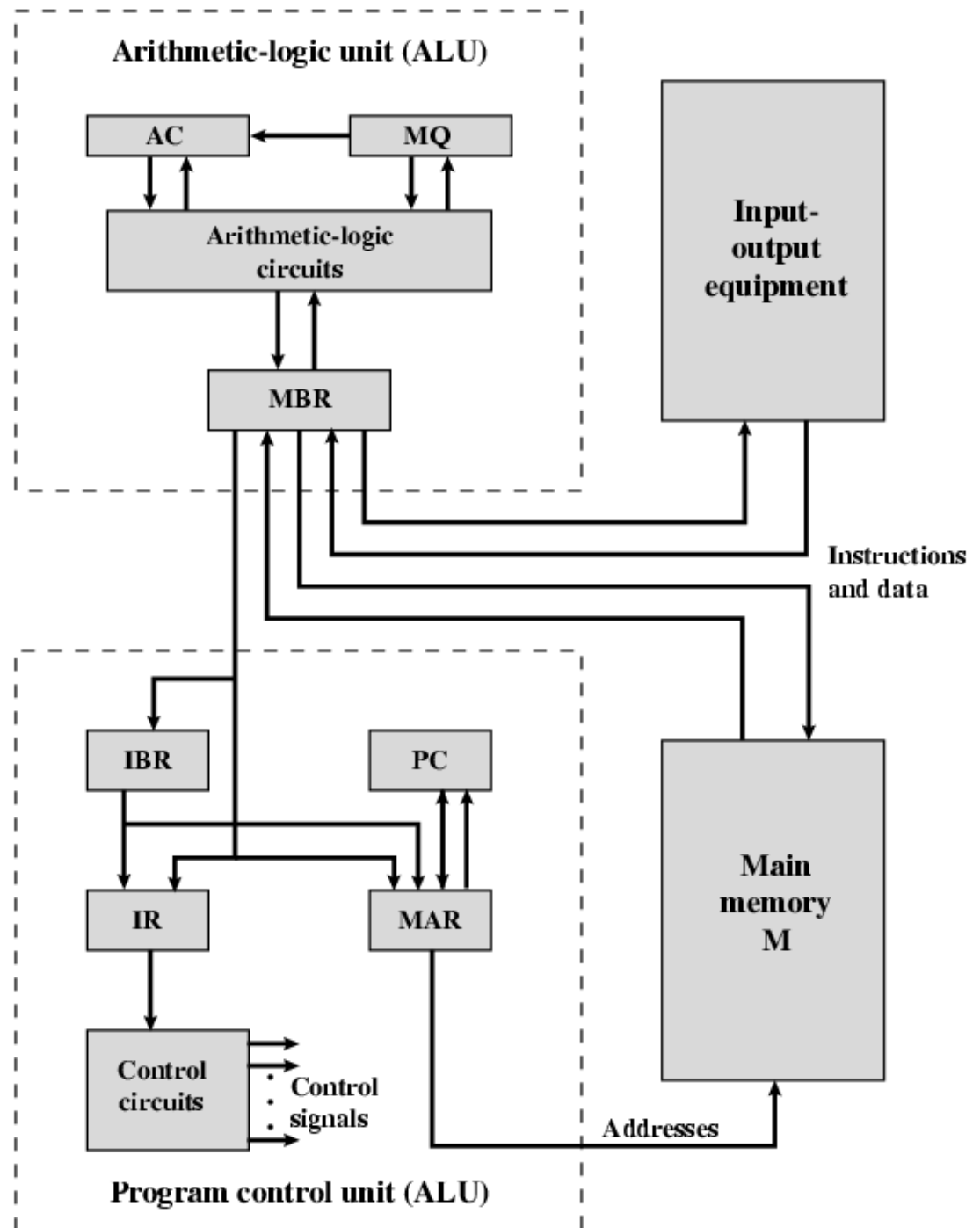
Cont'd...



- ✓ **Accumulator (AC) and multiplier quotient (MQ):**
Employed to hold temporarily operands and results of ALU operations.

For example, the result of multiplying two 40-bit numbers is an 80-bit number; **the most significant** 40 bits are stored in the AC and the **least significant** in the MQ.

Structure of IAS



IBM



- ❑ Punched-card processing equipment
- ❑ 1953 - the 701
 - ✓ IBM's first stored program computer
 - ✓ Scientific calculations
- ❑ Lead to 700/7000 series
- ❑ 1955 - the 702: Business applications

The Second Generation: Transistors



- ☐ Replaced vacuum tubes
- ☐ Smaller
- ☐ Cheaper
- ☐ Less heat dissipation
- ☐ Solid State device
- ☐ Made from Silicon (Sand)
- ☐ Invented 1947 at Bell Labs
- ☐ William Shockley et al.

Cont'd...



- ❑ Introduction of more complex arithmetic and logic units and control units,
- ❑ The use of high-level programming languages, and
- ❑ The provision of *system software* with the computer.

Third Generation: Integrated Circuits



- ❑ The separately manufactured components like resistor, capacitor, diode, and transistor are joined by wires or by printed circuit board (**PCB**) to form circuits. These circuits are called **discrete circuits**.
- ❑ An **IC** comprises a number of circuit components like resistors, transistor etc. They are interconnected **in a single small package** to perform the desired electronic function.

Scale of Integration



- ❑ The number of components fitted into a standard size IC represents its integration scale.
- ❑ It is classified as: **SSI, MSI, LSI, VLSI, ULSI**

Computer Generations Summary



Generation	Approximate Dates	Technology	Typical Speed (operations per second)
1	1946–1957	Vacuum tube	40,000
2	1958–1964	Transistor	200,000
3	1965–1971	Small- and medium-scale integration	1,000,000
4	1972–1977	Large-scale integration	10,000,000
5	1978–1991	Very-large-scale integration	100,000,000
6	1991–	Ultra-large-scale integration	1,000,000,000

Intel Microprocessors



❑ 1971 - **4004** *(The microprocessor was born!)*

- ✓ First microprocessor
- ✓ All CPU components on a single chip
- ✓ 4 bit

❑ Followed in 1972 by **8008**

- ✓ 8 bit
- ✓ Both designed for specific applications

❑ 1974 - **8080**

- ✓ Intel's first general purpose microprocessor

Cont'd...

(a) 1970s Processors

	4004	8008	8080	8086	8088
Introduced	1971	1972	1974	1978	1979
Clock speeds	108 kHz	108 kHz	2 MHz	5 MHz, 8 MHz, 10 MHz	5 MHz, 8 MHz
Bus width	4 bits	8 bits	8 bits	16 bits	8 bits
Number of transistors	2300	3500	6000	29,000	29,000
Feature size (μm)	10		6	3	6
Addressable memory	640 Bytes	16 kB	64 kB	1 MB	1 MB

(b) 1980s Processors

	80286	386TM DX	386TM SX	486TM DX CPU
Introduced	1982	1985	1988	1989
Clock speeds	6 MHz–12.5 MHz	16 MHz–33 MHz	16 MHz–33 MHz	25 MHz–50 MHz
Bus width	16 bits	32 bits	16 bits	32 bits
Number of transistors	134,000	275,000	275,000	1.2 million
Feature size (μm)	1.5	1	1	0.8–1
Addressable memory	16 MB	4 GB	16 MB	4 GB
Virtual memory	1 GB	64 TB	64 TB	64 TB
Cache	—	—	—	8 kB

Cont'd...

(c) 1990s Processors

	486TM SX	Pentium	Pentium Pro	Pentium II
Introduced	1991	1993	1995	1997
Clock speeds	16 MHz–33 MHz	60 MHz–166 MHz,	150 MHz–200 MHz	200 MHz–300 MHz
Bus width	32 bits	32 bits	64 bits	64 bits
Number of transistors	1.185 million	3.1 million	5.5 million	7.5 million
Feature size (µm)	1	0.8	0.6	0.35
Addressable memory	4 GB	4 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	8 kB	8 kB	512 kB L1 and 1 MB L2	512 kB L2

(d) Recent Processors

	Pentium III	Pentium 4	Core 2 Duo	Core i7 EE 990
Introduced	1999	2000	2006	2011
Clock speeds	450–660 MHz	1.3–1.8 GHz	1.06–1.2 GHz	3.5 GHz
Bus width	64 bits	64 bits	64 bits	64 bits
Number of transistors	9.5 million	42 million	167 million	1170 million
Feature size (nm)	250	180	65	32
Addressable memory	64 GB	64 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	512 kB L2	256 kB L2	2 MB L2	1.5 MB L2/12 MB L3

Multicore



- ❑ Multiple processors on single chip
 - ✓ Large shared cache
- ❑ Within a processor, increase in performance proportional to square root of increase in complexity
- ❑ If software can use multiple processors, doubling number of processors almost doubles performance
- ❑ So, use two simpler processors on the chip rather than one more complex processor
- ❑ With two processors, larger caches are justified
 - ✓ Power consumption of memory logic less than processing logic

The two well known processor families



- ❑ **Intel x86 architecture:** The x86 architecture is the most widely used for *non-embedded* computer systems. The x86 is essentially a complex instruction set computer (**CISC**).
- ❑ **ARM:** The ARM architecture is arguably the most widely used embedded processor, used in cell phones, iPods, remote sensor equipment, and many other devices. ARM is essentially a reduced instruction set computer (**RISC**).

Embedded Systems ARM

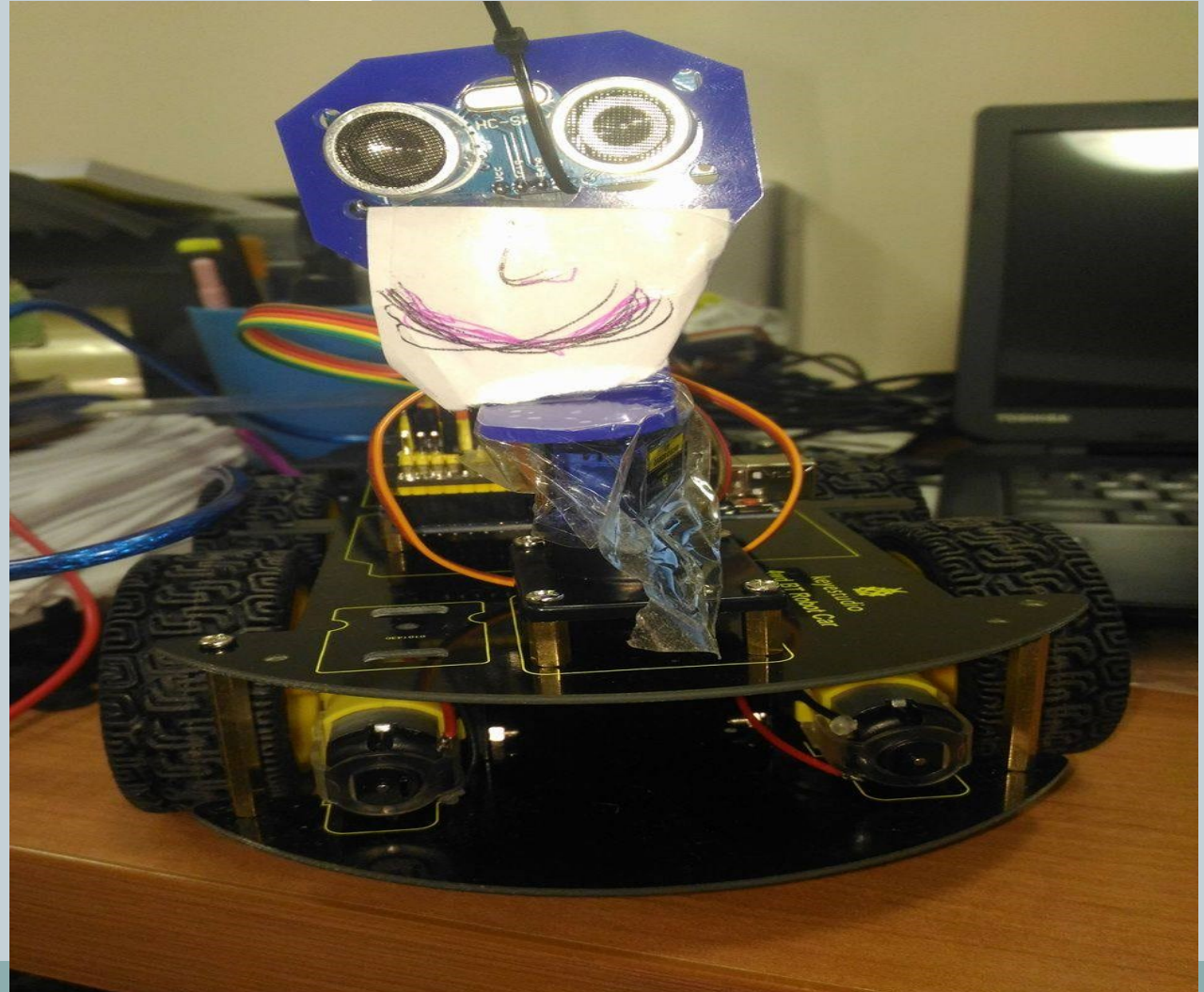


- ❑ ARM evolved from RISC design
- ❑ Used mainly in embedded systems
 - ✓ Used within product
 - ✓ Not general purpose computer
 - ✓ Dedicated function
 - ✓ E.g. **Anti-lock brakes** in car

Sample Embedded system



Collision Avoidance
robot



Cont'd...



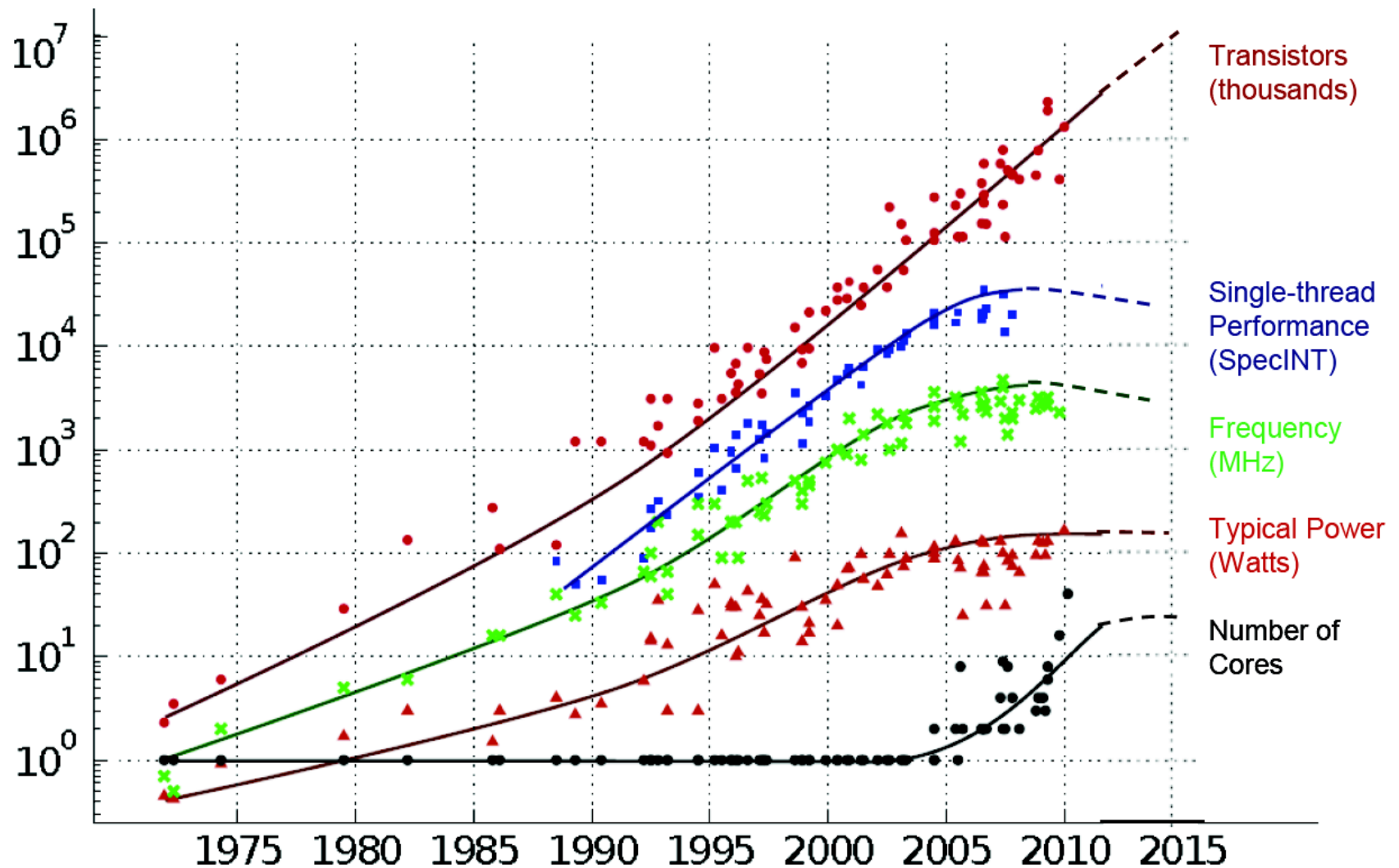
-
- The image shows a custom-built IoT device on a breadboard. The device includes a Raspberry Pi Zero, a 16x2 LCD display showing "Access Granted", a red LED, and various sensors. It is connected to a laptop via USB. The laptop screen shows a terminal window with a list of IP addresses and a file explorer window.

Moore's Law



- ❑ Increased density of components on chip
- ❑ **Gordon Moore** – co-founder of **Intel**
- ❑ Number of transistors on a chip will **double every year**
- ❑ Since 1970's development has slowed a little
 - Number of transistors doubles every 18 months
- ❑ Cost of a chip has remained almost unchanged
- ❑ Higher packing density means shorter electrical paths,
giving higher performance
- ❑ Smaller size gives increased flexibility
- ❑ Reduced power and cooling requirements
- ❑ Fewer interconnections increases reliability

35 YEARS OF MICROPROCESSOR TREND DATA



Original data collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond and C. Batten
Dotted line extrapolations by C. Moore

Cont'd...

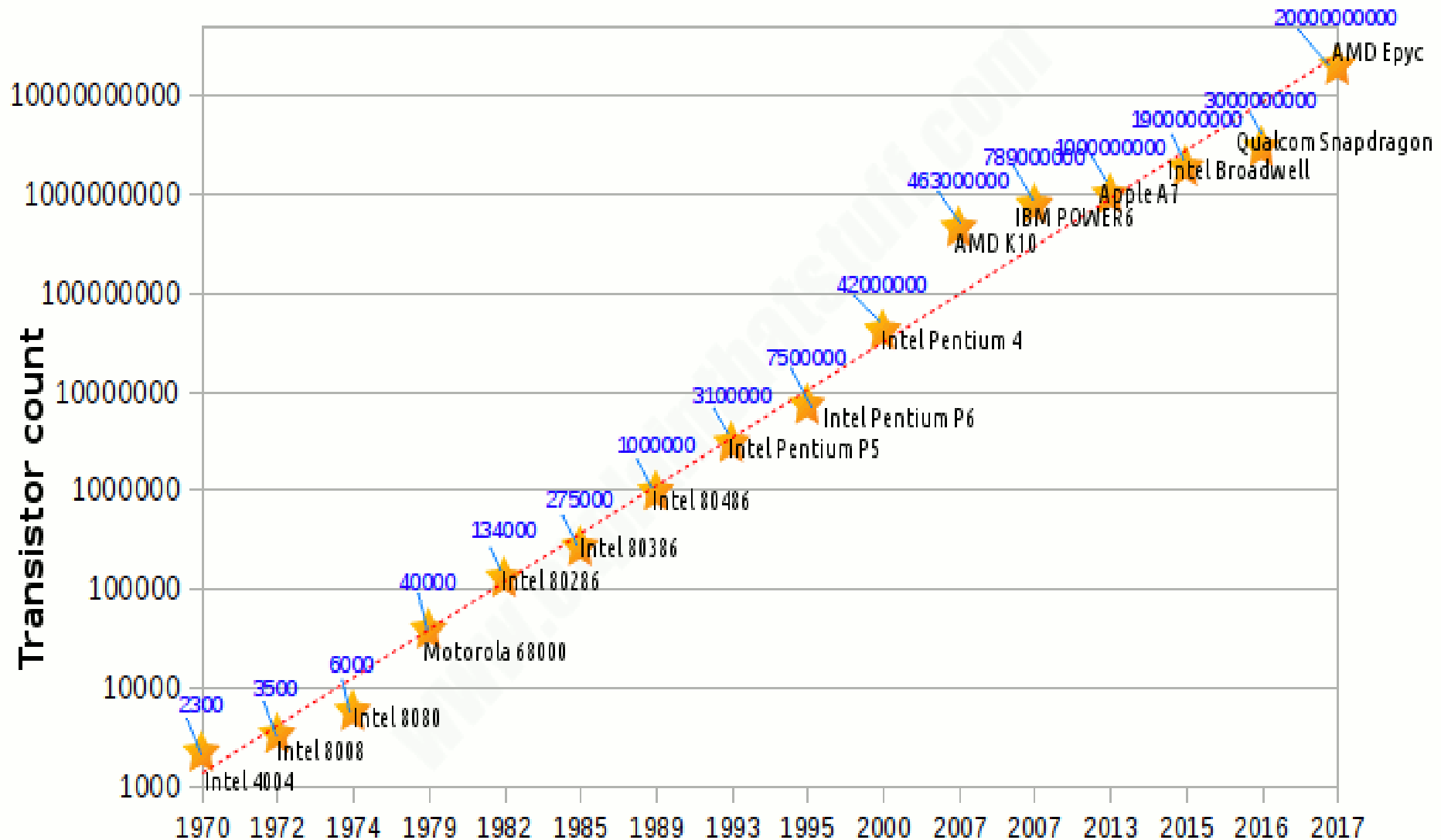




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50 Years of Moore's law

www.explainthatstuff.com



Questions?



Next Lecture



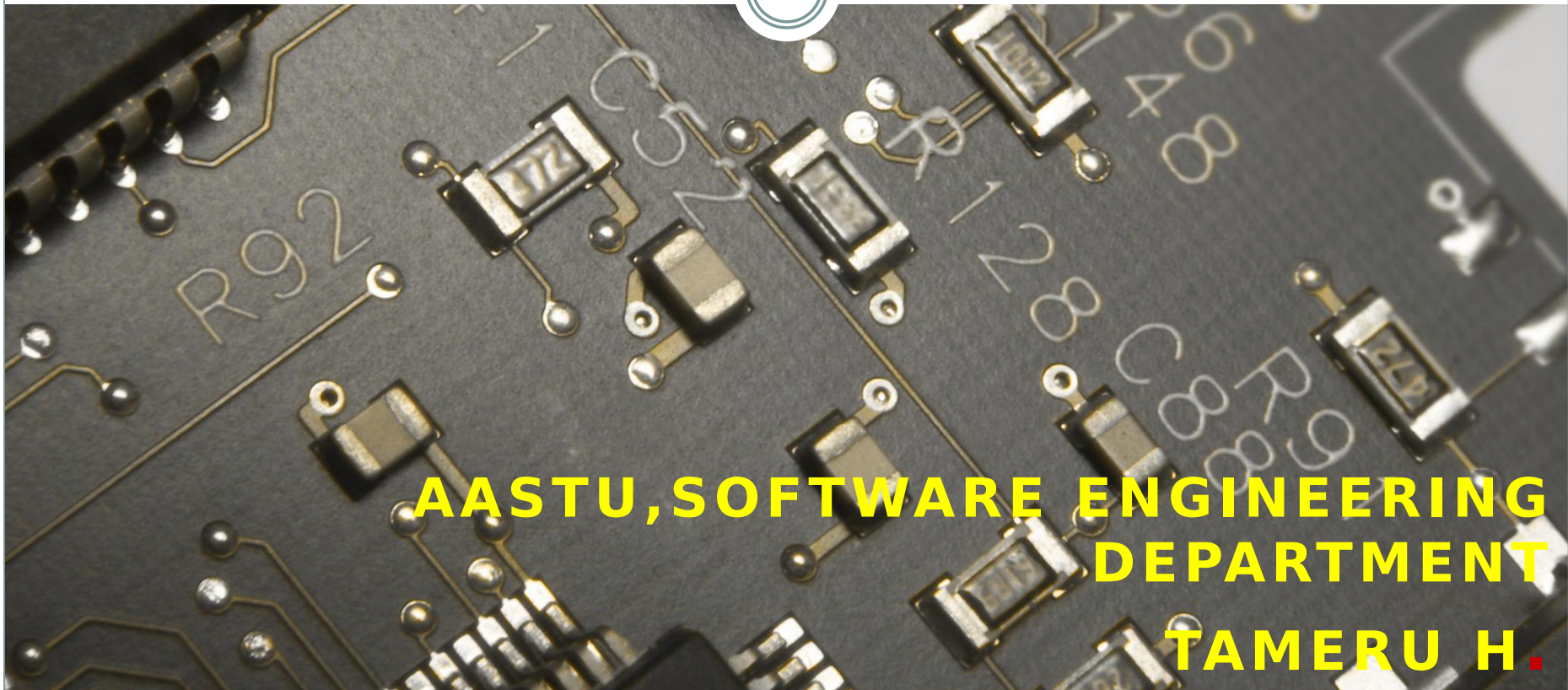
Chapter Two **Part-2**

Performance Assessment

Chapter Two

Part-2

Performance Assessment



Performance assessment



Key parameters to evaluate processor hardware:

- Performance,
- cost,
- size,
- security,
- reliability,
- power consumption

Cont'd...



When we say one
computer **has better**
performance than another,
what do we mean?



Cont'd...



What affects the
performance of your
application software?



Cont'd...



Application performance depends not just on the raw speed of the processor but also on:

- The instruction set,
- Choice of implementation language,
- Efficiency of the compiler, and
- Skill of the programming done to implement the application.

Clock Speed and Instructions per Second



Operations performed by a processor are governed by a system clock. Such as:

- fetching an instruction,
- decoding the instruction,
- performing an arithmetic operation and others

Cont'd...



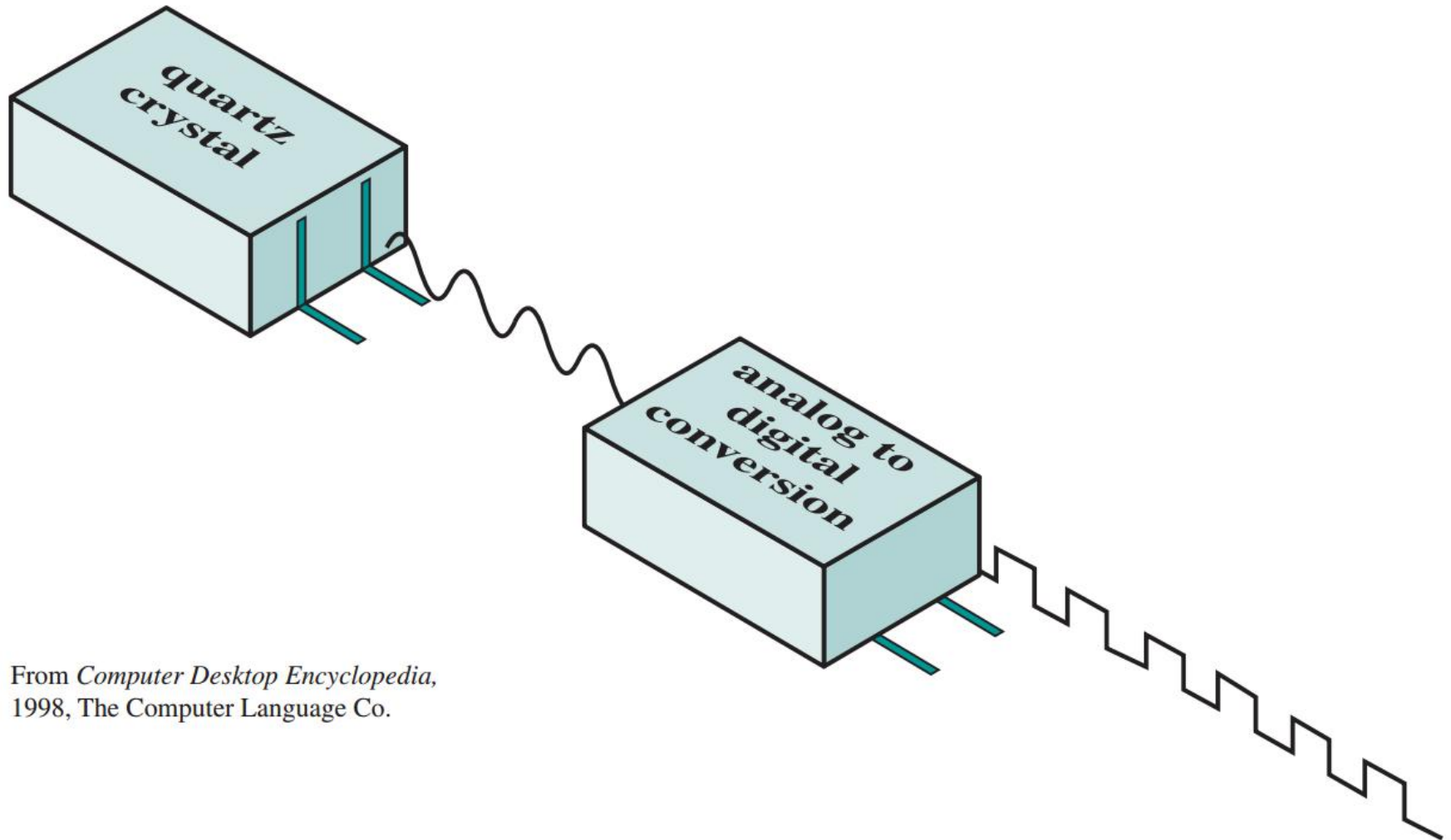
- All operations begin with the pulse of the clock.
- Thus, at the most fundamental level, the speed of a processor is **dictated** by the pulse frequency produced by the clock.
- Pulse is measured in cycles per second, or **Hertz (Hz)**.

Cont'd...



- All clock signals are generated by a **quartz crystal**, which generates a **constant signal** wave while power is applied.
- This wave is converted into a **digital voltage** pulse stream that is provided in a constant flow to the processor circuitry.

Cont'd...



From *Computer Desktop Encyclopedia*,
1998, The Computer Language Co.

Cont'd...



- The rate of pulses is known as the **clock rate**, or **clock speed**.
- One increment, or pulse, of the clock is referred to as a **clock cycle**, or a **clock tick**.
- The time between pulses is the **cycle time**.

Cont'd...



- The execution of an instruction involves a number of **discrete steps**.
- Thus, most instructions on most processors require **multiple clock** cycles to complete.
- Some instructions may take **only a few** cycles, while others require dozens.

Cont'd...



Thus, a straight comparison of clock speeds on different processors does not tell the whole story about performance.

Instruction Execution Rate



A processor is driven by :

- a clock with a constant frequency f or
- equivalently, a constant cycle time t , where $t = 1/f$.
- I_c , for a program as the number of machine instructions executed for that program until it runs to completion
- An important parameter is **the average cycles per instruction** (CPI) for a program.

Cont'd...



Let CPI_i be the number of cycles required for instruction type i and I_i be the number of executed instructions of type i for a given program.

Then we can calculate an overall CPI as follows:

$$CPI = \frac{\sum_{i=1}^n (CPI_i \times I_i)}{I_c}$$

Cont'd...



The **processor time** T needed to execute a given program can be expressed as

$$T = I_c \times CPI \times \tau$$

Cont'd...



A common measure of performance for a processor is the rate at which instructions are executed, expressed as millions of instructions per second (**MIPS**),

$$\text{MIPS rate} = \frac{I_c}{T \times 10^6} = \frac{f}{CPI \times 10^6}$$

Reading Assignment



Amdahl's law

Case study-1



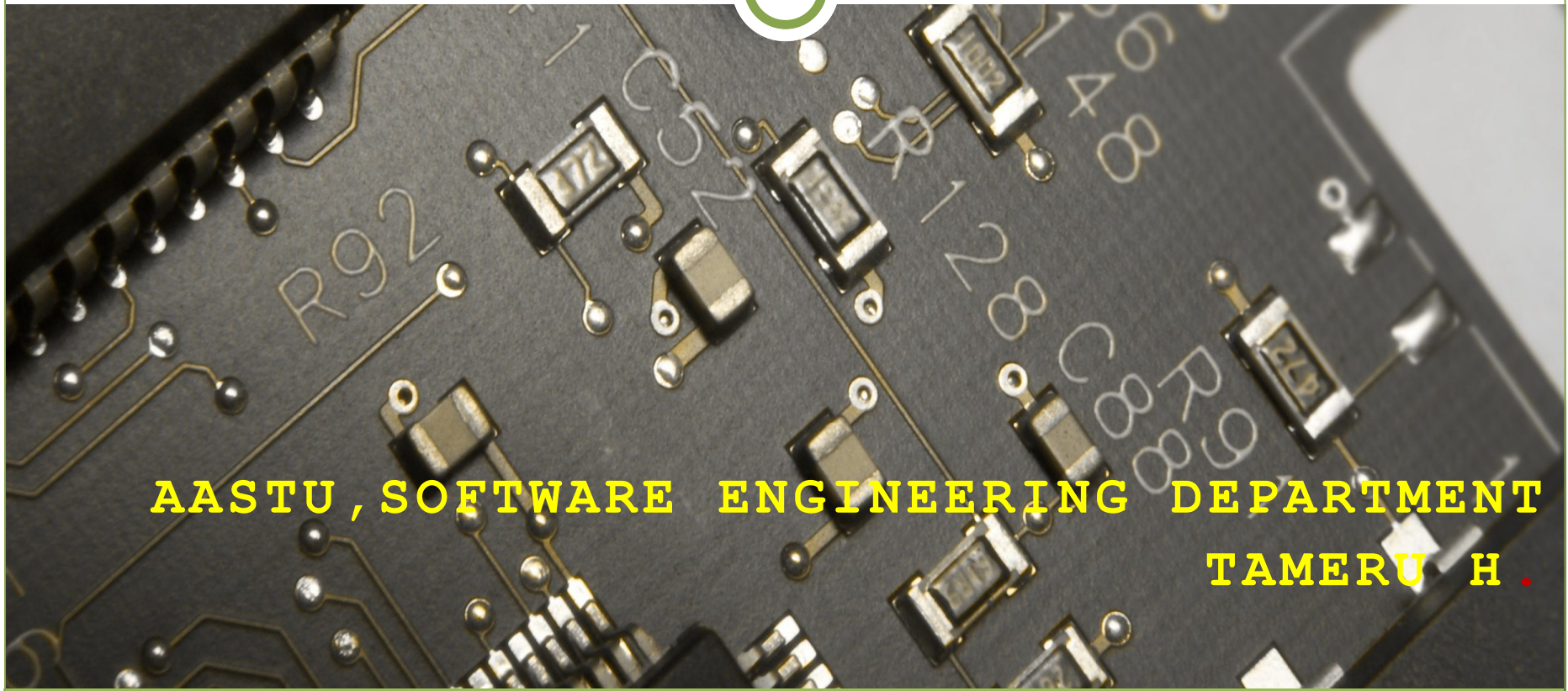
Our favorite program *runs in 10 seconds* on computer **A**, which has a **2 GHZ** clock. We are trying to help a computer designer build a computer, **B**, which will run this program in *6 seconds*. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing computer B *to require 1.2 times* as many clock cycles as computer A for this program. What clock rate should we tell the designer to target?

Case study-2



Suppose we have **two implementations** of the same instruction set architecture. Computer **A** has a clock cycle time of **250 ps** and a **CPI of 2.0** for some program, and computer **B** has a clock cycle time of **500 ps** and a **CPI of 1.2** for the same program.

Which computer is **faster** for this program and by how much?



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Amdahl's law



- Gene Amdahl [AMDA67]
- Potential speed up of program using multiple processors

For program running on single processor

- ✓ Fraction f of code infinitely parallelizable with no scheduling overhead
- ✓ Fraction $(1-f)$ of code inherently serial
- ✓ T is total execution time for program on single processor
- ✓ N is number of processors that fully exploit parallel portions of code

Cont'd..



$$\begin{aligned}\text{Speedup} &= \frac{\text{Time to execute program on a single processor}}{\text{Time to execute program on } N \text{ parallel processors}} \\ &= \frac{T(1 - f) + Tf}{T(1 - f) + \frac{Tf}{N}} = \frac{1}{(1 - f) + \frac{f}{N}}\end{aligned}$$

Cont'd...



Conclusions

- ✓ f small, parallel processors has little effect
- ✓ $N \rightarrow \infty$, speedup bound by $1/(1 - f)$

Diminishing returns for using more processors

Exercise 1



Consider 4 potential applications of the Amdahl's Law Formula:

1. 95% of a task/program/etc. is improved by 10%
2. 5% of a task/program/etc. is improved by 10X
3. 5% of a task/program/etc. is infinitely improved
4. 95% of a task/program/etc. is infinitely improved

For all 4 cases, what is the overall speedup of the task?

Exercise 2



Consider two different implementations, **M1** and **M2**, of the same instruction set. There are three classes of instructions (**A, B, and C**) in the instruction set. M1 has a clock rate of **80 MHz** and M2 has a clock rate of **100 MHz**. The average number of cycles for each instruction class and their frequencies (for a typical program) are as follows:

Instruction Class	Machine M1 – Cycles/Instruction Class	Machine M2 – Cycles/Instruction Class	Frequency
A	1	2	60%
B	2	3	30%
C	4	4	10%

(a) Calculate the average CPI for each machine, M1, and M2.

Cont'd..



(b) Calculate the average MIPS ratings for each machine, M1 and M2.

(c) Which machine has a smaller MIPS rating ?

Which individual instruction class CPI do you need to change, and by **how much**, to have this machine have the same or better performance as the machine with the higher MIPS rating (**you can only change the CPI for one of the instruction classes on the slower machine**)?

Exercise 3



Computer **A** has an overall **CPI of 1.3** and can be run at a clock rate of **600MHz**. Computer **B** has a **CPI of 2.5** and can be run at a clock rate of **750 Mhz**. We have a particular program we wish to run. When compiled for computer A, this program has exactly **100,000** instructions. How many instructions would the program need to have when compiled for Computer **B**, in order for the two computers to have exactly the same **execution time** for this program?

Exercise 4



The design team for a simple, single-issue processor is choosing between a pipelined or non-pipelined implementation. Here are some design parameters for the two possibilities:

Parameter	Pipelined Version	Non-Pipelined Version
Clock Rate	500MHz	350 MHz
CPI for ALU instructions	1	1
CPI for Control instructions	2	1
CPI for Memory instructions	2.7	1

Cont'd...



(a) For a program with **20% ALU instructions**, **10% control instructions** and **75% memory instructions**, which design will be faster? Give a quantitative CPI average for each case.

(b) For a program with **80% ALU instructions**, **10% control instructions** and **10% memory instructions**, which design will be faster? Give a quantitative CPI average for each case.